

# Nathan I. Wisnoski

Department of Biological Sciences, Mississippi State University

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## ACADEMIC APPOINTMENTS

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|-----------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Assistant Professor</b><br>Mississippi State University, Department of Biological Sciences | August 2022 – Present<br><i>Starkville, MS</i>   |
| <b>Postdoctoral Researcher</b><br>University of Wyoming                                       | July 2020 – August 2022<br><i>Laramie, WY</i>    |
| <b>Graduate Research &amp; Teaching Assistant</b><br>Indiana University                       | August 2014 – May 2020<br><i>Bloomington, IN</i> |

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Indiana University, Bloomington</b><br>Ph.D. in Biology – Evolution, Ecology, and Behavior<br>Minor in Environmental Studies, O’Neill School of Public and Environmental Affairs | 2014–2020 |
| <b>The University of Texas at Austin</b><br>B.S. in Biology – Ecology, Evolution, and Behavior<br>Minor in Business, McCombs School of Business                                     | 2009–2013 |

## PUBLICATIONS

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### *Preprints and Manuscripts Under Review*

1. Leibold, M. A., Barbier, M., Bittleston, L. S., Clark, A. T., Cuellar-Gempeler, C., D’Andrea, R., Frans, V. F., Khattar, G., Miller, Z., Peres-Neto, P., and **Wisnoski, N. I.** Linking pattern to process in metacommunities: challenges and opportunities. In review; preprint on EcoEvoRxiv. DOI: 10.32942/X2504P.
2. **Wisnoski, N. I.** Microbial biodiversity in river networks. *Aquatic Biodiversity: Theories, Approaches, and Future Directions*. Book chapter in press.
3. **Wisnoski, N. I.** Spatial microbial flows: hidden fluxes of detritus, diversity, and function in meta-ecosystems. In review; preprint on EcoEvoRxiv. DOI: 10.32942/X2BW46.

### *Published and In Press*

19. **Wisnoski, N. I.**, Szojka, M. C., Germain, R. M., Fukami, T., and Shoemaker, L. G. 2026. Dispersal kernels regulate environmental, biotic, and stochastic effects on the maintenance of metacommunity diversity. *Ecology* 107(9):e70484.
18. Davidson, J. L., McKnight, K. R., Szojka, M., Gannon, D., **Wisnoski, N. I.**, Werner, C. M., Liang, M., Seabloom, E. W., Ray, C., DeSiervo, M. H., and Shoemaker, L. G. 2025. Effects of disturbance and fertilization on plant community synchrony, biodiversity, and stability through succession. *Ecology Letters* 28(4):e70052.
17. Eggers, J. M. von, **Wisnoski, N. I.**, Calder, W. J., Capo, E., Groff, D. V., Krist, A. C., and Shuman, B. 2024. Environmental filtering governs consistent vertical zonation in sedimentary microbial communities across disconnected mountain lakes. *Environmental Microbiology* 26(3):e16607.
16. Mobilian, C., **Wisnoski, N. I.**, Lennon, J. T., Alber, M., Widney, S., and Craft, C. B. 2023. Differential effects of press vs. pulse seawater intrusion on microbial communities of a tidal freshwater marsh. *Limnology and Oceanography Letters* 8(1):151–161.

15. **Wisnoski, N. I.**, Andrade, R., Castorani, M. C. N., Catano, C. P., Compagnoni, A., Lamy, T., Lany, N. K., Marazzi, L., Record, S., Smith, A. C., Swan, C. M., Tonkin, J. D., Voelker, N. M., Zarnetske, P. L., and Sokol, E. R. 2023. Diversity–stability relationships become decoupled across spatial scales: a synthesis of organism and ecosystem types. *Ecology* 104(9):e4136.
14. **Wisnoski, N. I.** and Lennon, J. T. 2023. Scaling up and down: movement ecology for microorganisms. *Trends in Microbiology* 31(3):242–253.
13. **Wisnoski, N. I.** and Shoemaker, L. G. 2022. Seed banks alter metacommunity diversity: the interactive effects of competition, dispersal, and dormancy. *Ecology Letters* 25(4):740–753.
12. Graham, E. B., Averill, C., Bond-Lamberty, B., Knelman, J. E., Krause, S., Peralta, A. L., Shade, A., Smith, A. P., Cheng, S., Fanin, N., Freund, C., Garcia, P. E., Gibbons, S. M., Van Goethem, M. W., Guebila, M. B., Kemppinen, J., Nowicki, R., Pausas, J. G., Reed, S., Rocca, J., Sengupta, A., Sihi, D., Simonin, M., Słowiński, M., Spawn, S., Sutherland, I., Tonkin, J., **Wisnoski, N. I.**, Zipper, S. C., and Contributor Consortium. 2021. Toward a unifying framework of disturbance ecology through crowdsourced science. *Frontiers in Ecology and Evolution* 9:76.
11. Lamy, T., **Wisnoski, N. I.**, Andrade, R., Castorani, M. C. N., Compagnoni, A., Lany, N., Marazzi, L., Record, S., Swan, C. M., Tonkin, J. D., Voelker, N., Wang, S., Zarnetske, P. L., and Sokol, E. R. 2021. The dual nature of metacommunity variability. *Oikos* 130(12):2078–2092. Editor's Choice.
10. Record, S., Voelker, N. M., Zarnetske, P. L., **Wisnoski, N. I.**, Tonkin, J. D., Swan, C. M., Marazzi, L., Lany, N., Lamy, T., Compagnoni, A., Castorani, M. C. N., Andrade, R., and Sokol, E. R. 2021. Novel insights to be gained from applying metacommunity theory to long-term, spatially replicated biodiversity data. *Frontiers in Ecology and Evolution* 8:479.
9. Shoemaker, L. G., Walter, J. A., Gherardi, L. A., DeSiervo, M. H., and **Wisnoski, N. I.** 2021. Writing mathematical ecology: a guide for authors and readers. *Ecosphere* 12(8):e03701.
8. **Wisnoski, N. I.** and Lennon, J. T. 2021. Microbial community assembly in a multi-layer dendritic metacommunity. *Oecologia* 195:13–24. Highlighted Student Research.
7. **Wisnoski, N. I.** and Lennon, J. T. 2021. Stabilizing role of seed banks and the maintenance of bacterial diversity. *Ecology Letters* 24(11):2328–2338.
6. Mueller, E. A., **Wisnoski, N. I.**, Peralta, A. L., and Lennon, J. T. 2020. Microbial rescue effects: how microbiomes can save hosts from extinction. *Functional Ecology* 34(10):2055–2064.
5. **Wisnoski, N. I.**, Muscarella, M. E., Larsen, M. L., Peralta, A. L., and Lennon, J. T. 2020. Metabolic insight into bacterial community assembly across ecosystem boundaries. *Ecology* 101(4):e02968.
4. Ward, A. S., Kurz, M. J., Schmadel, N. M., Knapp, J. L. A., Blaen, P. J., Harman, C. J., Drummond, J., Hannah, D. M., Krause, S., Li, A., Martí, E., Milner, A., Miller, M., Neil, K., Plont, S., Packman, A. I., **Wisnoski, N. I.**, Wondzell, S. M., and Zarnetske, J. P. 2019. Solute transport and transformation in an intermittent, headwater mountain stream with diurnal discharge fluctuations. *Water* 11(11):2208.
3. Ward, A. S., Wondzell, S. M., Schmadel, N. M., Herzog, S., Zarnetske, J. P., Baranov, V., Blaen, P. J., Brekenfeld, N., Chu, R., Derelle, R., Drummond, J., Fleckenstein, J. H., Garayburu-Caruso, V., Graham, E., Hannah, D., Harman, C. J., Hixson, J., Knapp, J. L. A., Krause, S., Kurz, M. J., Lewandowski, J., Li, A., Martí, E., Miller, M., Milner, A. M., Neil, K., Orsini, L., Packman, A. I., Plont, S., Renteria, L., Roche, K., Royer, T., Segura, C., Stegen, J., Toyoda, J., Wells, J., and **Wisnoski, N. I.** 2019. Spatial and temporal variation in river corridor exchange across a 5th-order mountain stream network. *Hydrology and Earth System Sciences* 23:5199–5225.
2. Ward, A. S., Zarnetske, J. P., Baranov, V., Blaen, P. J., Brekenfeld, N., Chu, R., Derelle, R., Drummond, J., Fleckenstein, J. H., Garayburu-Caruso, V., Graham, E., Hannah, D., Harman, C. J., Herzog, S., Hixson, J., Knapp, J. L. A., Krause, S., Kurz, M. J., Lewandowski, J., Li, A., Martí, E., Miller, M., Milner, A. M., Neil, K., Orsini, L., Packman, A. I., Plont, S., Renteria, L., Roche, K., Royer, T., Schmadel, N. M., Segura, C., Stegen, J., Toyoda, J., Wells, J., **Wisnoski, N. I.**, and Wondzell, S. M. 2019. Co-located contemporaneous mapping of morphological, hydrological, chemical, and biological conditions in a 5th-order mountain stream network, Oregon, USA. *Earth System Science Data* 11:1567–1581.

1. **Wisnoski, N. I.**, Leibold, M. A., and Lennon, J. T. 2019. Dormancy in metacommunities. *The American Naturalist* 194(2):135–151.

### Book Reviews

1. **Wisnoski, N. I.** and Lennon, J. T. 2016. Review of “Principles of Microbial Diversity” by James W. Brown. *The Quarterly Review of Biology* 91(1):98–99.

## GRANTS AND FUNDING

### Proposals Under Review

|         |             |                                                                                    |
|---------|-------------|------------------------------------------------------------------------------------|
| Pending | \$1,196,417 | NSF CAREER. PI: <b>N.I. Wisnoski</b> .                                             |
| Pending | \$7,500,000 | FDA Animal and Veterinary Innovation Centers (AVIC). Co-PI: <b>N.I. Wisnoski</b> . |

### Awarded Funding

Total awarded: \$1,106,783

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|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027–2028 | \$299,738 | NSF EPSCoR Research Fellows. <i>Linking microbial metabolic networks to ecosystem-scale fluxes in coastal mangroves with a multi-omics and ML/AI approach</i> . PI: <b>N.I. Wisnoski</b> .                                                                                |
| 2026–2028 | \$45,000  | Mississippi State University. <i>Using remote sensing to monitor ecosystem health, productivity, and disturbance in southern pine forest landscapes</i> . PI: R. Bhattarai; Co-PI: <b>N.I. Wisnoski</b> .                                                                 |
| 2023–2026 | \$614,757 | Mississippi Based RESTORE Act Center of Excellence (MBRACE). <i>Transport and fate of bacterial communities in Mississippi coastal ecosystems</i> . PI: <b>N.I. Wisnoski</b> .                                                                                            |
| 2021–2022 | \$65,888  | Wyoming EPSCoR: Microbial Ecology Collaborative. <i>Microbial community assembly in dendritic networks: disentangling the roles of local environmental heterogeneity, stochasticity, and dispersal</i> . PIs: <b>N.I. Wisnoski</b> and S.E. Albeke.                       |
| 2016–2018 | \$76,000  | NSF LTER Network Communications Office (NCEAS). <i>A synthesis to identify how metacommunity dynamics mediate community responses to disturbance across the ecosystems represented in the LTER network</i> . PI: E.R. Sokol; Co-PIs: C.M. Swan and <b>N.I. Wisnoski</b> . |
| 2015      | \$5,400   | Indiana University. Sustainability Research Development Grant.                                                                                                                                                                                                            |

## FELLOWSHIPS AND AWARDS

|                                                                                 |      |
|---------------------------------------------------------------------------------|------|
| Downer Undergraduate Research Award, First Place Mentor Award                   | 2024 |
| Louise Constable Hoover Fellowship, IU Biology, \$2,000                         | 2019 |
| Travel Award, Association for the Sciences of Limnology and Oceanography, \$606 | 2018 |
| Travel Award, ESA Microbial Ecology Section, \$600                              | 2017 |
| George W. Brackenridge Fellowship, IU Biology, \$2,000                          | 2016 |
| Travel Award, International Society for Microbial Ecology, €300                 | 2016 |
| Travel Award, Honorable Mention, ESA Microbial Ecology Section, \$150           | 2016 |
| Departmental Research Recruitment Fellowship, IU Biology,                       | 2014 |

## INVITED PRESENTATIONS

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- 2024 Invited speaker, American Society of Microbiology Tennessee-Kentucky and South-Central Joint Branch Meeting, Memphis, TN
- 2024 Invited speaker, “Organized Oral Session: Linking pattern to process in (meta)community assembly”, Ecological Society of America, Long Beach, CA
- 2023 Invited speaker, “Workshop: Critical phenomena and challenges emerging from dormancy”, Goethe Universität, Frankfurt, Germany
- 2022 Seminar, Swiss Federal Institute of Aquatic Science and Technology (Eawag), Department of Aquatic Ecology
- 2022 Seminar, Mississippi State University, Department of Biochemistry, Molecular Biology, Entomology & Plant Pathology
- 2022 Invited speaker, “Workshop: Impacts of dormancy and latency on host-parasite dynamics”, Universität Bonn, Hausdorff Research Institute for Mathematics (HIM), Bonn, Germany
- 2021 Seminar, Mississippi State University, Department of Biological Sciences
- 2020 Seminar, University of Wyoming, Department of Botany

## CONTRIBUTED PRESENTATIONS

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Note: \* = Presenter

- Patalinghug, J. M. R.\*, Castañeda-Moya, E., Kominoski, J. S., Travieso, R., and **Wisnoski, N. I.** 2026. *Mangroves, microbes, and metacommunities along Shark River Estuary*. Florida Coastal Everglades LTER All Scientists’ Meeting. Miami, FL.
- Sharna, S. N.\*, Wise, D. J., Griffin, M. J., and **Wisnoski, N. I.** 2025. *Vaccination effects on microbiome assembly in channel catfish and pond environments*. American Society for Microbiology - South Central Branch Annual Meeting. Starkville, MS.
- Patalinghug, J. M. R.\* and **Wisnoski, N. I.** 2025. *Microbial ecology of mangroves: a metacommunity perspective*. American Society for Microbiology - South Central Branch Annual Meeting. Starkville, MS.
- Patalinghug, J. M. R.\* and **Wisnoski, N. I.** 2025. *Microbial ecology of mangroves: a metacommunity perspective*. Florida Coastal Everglades LTER All Scientists’ Meeting. Miami, FL.
- Wisnoski, N. I.\*** 2025. *Dispersal-dormancy covariation and the maintenance of biodiversity*. Ecological Society of America. Baltimore, MD.
- Singh, T.\* and **Wisnoski, N. I.** 2024. *The fate of terrestrial-derived bacteria in aquatic ecosystems* (poster). Ecological Society of America. Long Beach, CA.
- Davidson, J. L.\*, McKnight, K., DeSiervo, M. H., Werner, C. M., Szojka, M. C., **Wisnoski, N. I.**, Seabloom, E. W., and Shoemaker, L. G. 2022. *Community synchrony, stability, and biodiversity responses to global change: evaluating short-term versus multi-decadal responses*. Ecological Society of America Annual Meeting. Montreal, QC, Canada.
- Eggers, J. M. von\*, Groff, D., **Wisnoski, N. I.**, Calder, W. J., Krist, A., and Shuman, B. 2022. *Controls on biogeography and assembly of lake sedimentary microbial communities*. Joint Aquatic Sciences Meeting. Grand Rapids, MI.
- Wisnoski, N. I.\***, Muscarella, M. E., Larsen, M. L., Peralta, A. L., and Lennon, J. T. 2021. *Metabolic insight into bacterial community assembly across ecosystem boundaries*. Ecological Society of America Annual Meeting. Long Beach, CA (Virtual).
- Wisnoski, N. I.** and Sokol, E. R.\*. 2021. *Diversity-stability relationships across spatial scales: aggregate and compositional variability correlate with different dimensions of metacommunity diversity*. Ecological Society of America Annual Meeting. Long Beach, CA (Virtual).
- Wisnoski, N. I.\***, Sokol, E. R., Andrade, R., Castorani, M. C. N., Catano, C. P., Compagnoni, A., Lamy, T., Lany, N. K., Marazzi, L., Record, S., Smith, A. C., Swan, C. M., Tonkin, J. D., Voelker, N. M., and Zarnetske, P. L. 2019. *Patterns and drivers of stability in long-term metacommunity data*. Ecological Society of America Annual Meeting. Louisville, KY.

- Sokol, E. R.\*, **Wisnoski, N. I.**, and Swan, C. M. 2019. *Insights from the synthesis of long-term biodiversity data: resources and tools available to community ecologists*. Ecological Society of America Annual Meeting. Louisville, KY.
- Wisnoski, N. I.\***, Leibold, M. A., and Lennon, J. T. 2019. *Dormancy in metacommunities: when can temporal dispersal maintain diversity in variable landscapes?* Society for Freshwater Science Annual Meeting. Salt Lake City, UT.
- Ward, A. S.\*, Harman, C. J., Schmadel, N. M., Kurz, M. J., Blaen, P., Wondzell, S. M., Drummond, J. D., Hannah, D. M., Knapp, J. L., Krause, S., Li, A., Martí, E. R., Miller, M., Milner, A., Neil, K., Plont, S., Roche, K. R., Packman, A. I., **Wisnoski, N. I.**, and Zarnetske, J. P. 2018. *How do evapotranspiration-driven discharge fluctuations alter reach-scale ecosystem function?* American Geophysical Union, Fall Meeting. Washington, D.C.
- Ward, A. S.\*, Herzog, S., Wondzell, S. M., Schmadel, N. M., Blaen, P., Drummond, J. D., Hannah, D. M., Harman, C. J., Knapp, J. L., Krause, S., Kurz, M. J., Li, A., Martí, E., Miller, M., Milner, A., Neil, K., Plont, S., Roche, K. R., Packman, A. I., **Wisnoski, N. I.**, and Zarnetske, J. P. 2018. *Spatial and temporal relationships between hydrologic forcing, geologic setting, and river corridor exchange in a mountain stream network*. American Geophysical Union, Fall Meeting. Washington, D.C.
- Ward, A. S.\*, Herzog, S., Wondzell, S. M., Schmadel, N., Blaen, P., Drummond, J., Hannah, D. M., Harman, C. J., Knapp, J., Krause, S., Kurz, M. J., Li, A., Martí, E., Miller, M., Milner, A., Neil, K., Plont, S., Roche, K., Packman, A. I., **Wisnoski, N. I.**, and Zarnetske, J. 2018. *How do hydrologic forcing and geologic setting control river corridor exchange in a 5th order mountain stream network?* Geological Society of America Annual Meeting. Indianapolis, IN.
- Wisnoski, N. I.\*** and Lennon, J. T. 2018. *Contribution of “seed banks” to bacterioplankton community dynamics*. Society for Freshwater Science Annual Meeting. Detroit, MI.
- Sokol, E. R.\*, **Wisnoski, N. I.**, and Swan, C. M. 2018. *Using long-term data to understand when metacommunities respond to disturbance*. Ecological Society of America Annual Meeting. New Orleans, LA.
- Wisnoski, N. I.\***, Muscarella, M. E., and Lennon, J. T. 2018. *Dispersal and dormancy across ecosystem boundaries*. Association for the Sciences of Limnology and Oceanography. Victoria, BC, Canada.
- Wisnoski, N. I.\*** and Lennon, J. T. 2017. *Microbial community assembly in dendritic metacommunities*. Ecological Society of America Annual Meeting. Portland, OR.
- Sokol, E. R.\*, **Wisnoski, N. I.**, Swan, C. M., Andrade, R., Bateman, H. L., Hope, A. G., Kominoski, J., Lany, N. K., Marazzi, L., Presley, S. J., Rassweiler, A., Record, S., Willig, M. R., and Zarnetske, P. L. 2017. *The role of long-term ecological research programs for testing metacommunity theory and understanding biodiversity patterns*. Ecological Society of America Annual Meeting. Portland, OR.
- Voelker, N. M.\*, Sokol, E. R., **Wisnoski, N. I.**, Swan, C. M., Lamy, T., Castorani, M. C. N., Marazzi, L., Compagnoni, A., Blanchard, J. R., Andrade, R., and Lany, N. K. 2017. *Evaluating the link between metacommunity stability and environmental variability across trophic groups represented at LTER sites*. Ecological Society of America Annual Meeting. Portland, OR.
- Wisnoski, N. I.\*** and Lennon, J. T. 2016. *Community assembly processes differ between surface water and sediment-associated communities in stream networks*. Ecological Society of America Annual Meeting. Fort Lauderdale, FL.
- Wisnoski, N. I.\*** and Lennon, J. T. 2016. *Local and regional processes in stream microbial community assembly* (poster). International Symposium on Microbial Ecology (ISME 16). Montreal, QC, Canada.
- Wisnoski, N. I.\***, Ward, A. S., and Lennon, J. T. 2015. *Bacterial metacommunity structure across a stream network* (poster). LTER All Scientists Meeting. Estes Park, CO.

## WORKSHOPS AND SYMPOSIA

- |      |                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Invited participant, “Dormancy working group”, Yale University, New Haven, CT                                                                  |
| 2025 | Invited participant, “MSU Aquatic Antimicrobial Resistance Workshop”, FARO Reference Center on AMR and Aquaculture Biosecurity, Starkville, MS |

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| 2025      | Co-organizer, “Organized Oral Session: Evolutionary ecology of dormancy in communities”, Ecological Society of America, Baltimore, MD                              |
| 2025      | Invited participant and microbial lead, “River Corridor Critical Zone RCN Workshop”, HJ Andrews Experimental Forest LTER, Oregon                                   |
| 2022      | Invited working group participant, “Workshop: Merging Statistical Theory and Analyses at the Interface of Microbial and ‘Macrobial’ Ecology”, Montreal, QC, Canada |
| 2018      | Lead organizer, “Synthesizing long-term community data: questions, challenges, and advances”, LTER All Scientists Meeting, Pacific Grove, CA                       |
| 2016–2023 | Co-organizer, “LTER metacommunities working group”, NSF LTER Network Communications Office, NCEAS, Santa Barbara, CA                                               |

## TEACHING

| <i>Dates</i>    | <i>Role</i>          | <i>University</i>            | <i>Course</i>                                                         |
|-----------------|----------------------|------------------------------|-----------------------------------------------------------------------|
| 2023–2026       | Instructor           | Mississippi State University | BIO 4993/6993: Community Ecology                                      |
| 2023–2026       | Instructor           | Mississippi State University | BIO 3104: Ecology                                                     |
| 2016; 2018–2020 | Associate Instructor | Indiana University           | BIOL-L 111: Foundations of Biology: Diversity, Evolution, and Ecology |
| 2014; 2017–2019 | Associate Instructor | Indiana University           | BIOL-L 113: Biology Laboratory                                        |
| 2017            | Co-Instructor        | Indiana University           | BIOL-Z 620: Quantitative Biodiversity                                 |
| 2014            | Grader               | University of Texas          | BIO 364: Microbial Ecology                                            |
| 2013            | Teaching Assistant   | University of Texas          | SSC 328M: Biostatistics                                               |

## ACADEMIC ADVISEES

### *Postdoctoral Researcher*

- Durga Purushotham Mahesh Chinthalapudi (2025–present)

### *Ph.D. Students*

- Jenny Marie Patalinghug (2024–present)
- Sadia Nusrat Sharna (2024–present)
- Brenda Wu (2026–present)

## SERVICE

### *Mentorship*

|                         |                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graduate committees     | Mississippi State University: Tristan Henderson, Garrett Timmons, Taylor Bishop, Shamim Ahmed, Pranoti Giram, Caleb Perniciaro, Paul Perez, Eric Ahiadu, Niranjan Paudel, and Prakriti Pathak                  |
| Undergraduates          | Mississippi State University: Tajinder Singh and Portia Mobley<br>Indiana University: Luke Pryke, Mollie Carrison, Jaylen Beatty (Summer REU), Mary Wallace (Summer REU), and SydneyEllen Gooding (Summer REU) |
| High school researchers | Indiana University: Dakayla Calhoun, Samuel Iwu, and Ian Schowe                                                                                                                                                |

### *Grant Review Panels*

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|------|----------------------------------------|
| 2022 | LTER Network, Synthesis Working Groups |
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*Ad Hoc Grant Review*

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|------------|----------------------------------------------------------------|
| 2025       | John R. Evans Leaders Fund (JELF)                              |
| 2024–2025  | Swiss National Science Foundation                              |
| 2020, 2022 | National Science Foundation, Division of Environmental Biology |

*Ad Hoc Peer Review*

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journals | American Naturalist, Applied and Environmental Microbiology, Aquatic Ecology, Austral Ecology, Biogeosciences, Biology Letters, BioScience, Current Biology, Ecography, Ecological Applications, Ecological Indicators, Ecology, Ecology Letters, Ecosphere, Ecosystem Health and Sustainability, Environmental Microbiology, FEMS Microbial Ecology, Freshwater Science, Global Ecology and Biogeography, The ISME Journal, Journal of Biogeography, Journal of Ecology, Limnology and Oceanography, Methods in Ecology and Evolution, Microbial Ecology, mSystems, Nature Communications, New Phytologist, npj Biofilms, Oikos, Philosophical Transactions of the Royal Society B, PNAS Nexus, Proceedings of the National Academy of Sciences, Scientific Reports, Water Resources Research |
| Books    | Kirchman (2018), <i>Processes in Microbial Ecology</i> (2nd ed.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Theses   | University of Oulu (Ph.D., 2023)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

*Professional Societies*

|      |                                                                                                                                                                        |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Organizing Committee, ASM-South Central Branch Annual Meeting, Starkville, MS                                                                                          |
| 2017 | Moderator, Organized Oral Session: <i>Advancing ecological theory through synthesis of long-term ecological research</i> , Ecological Society of America, Portland, OR |

*Department*

|              |                                                                                              |
|--------------|----------------------------------------------------------------------------------------------|
| 2025–2026    | Faculty Search Committee, Department of Biological Sciences, Mississippi State University    |
| 2025–2026    | Biology Seminar Coordinator, Department of Biological Sciences, Mississippi State University |
| 2024–present | Graduate Committee, Department of Biological Sciences, Mississippi State University          |
| 2023–2024    | Faculty Search Committee, Department of Biological Sciences, Mississippi State University    |
| 2022–present | Zernickow Award Committee, Department of Biological Sciences, Mississippi State University   |
| 2014–2020    | Graduate Recruiting Weekend, Department of Biology, Indiana University                       |
| 2015–2016    | EcoLunch Co-Organizer, Indiana University                                                    |
| 2015         | Organizer, Metacommunity Reading Group, Indiana University                                   |

*Community Outreach*

|           |                                              |
|-----------|----------------------------------------------|
| 2017–2019 | Coordinator, High School Riverwatch Sampling |
|-----------|----------------------------------------------|